

THE THREE-RING THEOREM

The composite discriminant group, bound by three proven laws — and the prime the rings could not pull apart

Rafael Amichis Luengo · Madrid · tretoef@gmail.com · [github.com/tretoef-estrella](https://github.com/tretoef/estrella) · 8 June 2026 *Companion to THE WATERMARK THEOREM, THE DOUBLE LADDER THEOREM, THE SWEET LIE THEOREM, THE BEND THEOREM and THE WILD TOOTH THEOREM (same repository). Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.*

Three theorems of this collection each settle one face of the Fermat surface discriminant at prime degree: the **Watermark** fixes its *order*, the **Double Ladder** fixes its *group structure*, the **Bend** fixes how a small prime dividing the degree *shifts* the count. Composite degree is where the three must meet. This document binds them into a single law for the order- m^2 factor count — the count that decides the whole elementary-divisor profile — and reports it measured firm across eight stations, blind-confirmed twice. But the binding came at the price of a long hunt, and the hunt returned a result sharper than the law: a question this campaign asked of the composite group again and again, in five different disguises, and which the group answered, every time, with the same refusal. The bend $b(p)$ does not live in a corner of the group that can be cut away and named. It lives inside the primitive heart of the surface, woven into the order- m characters, inseparable. Five routes tried to pull it loose; five routes died with data; and the obituary is the theorem. *The teeth of the Bend are not glued onto the surface — they grew from its bone.*

0. STATEMENT

Let S_m be the Fermat surface of composite degree m , $NS(S_m)$ its Néron-Severi lattice, and let b_p denote the number of order- m^2 cyclic factors of the discriminant group $NS(S_m)^*/NS(S_m)$ carried at the prime $p \mid m$ — the composite continuation of the Double Ladder's order- m^2 count.

Theorem (the Three-Ring).

- **[The count law — FIRM, measured 8/8]** For squarefree composite m and each prime $p \mid m$, $b_p = (3m - 16) + b(p)$ — the Double Ladder base $3m - 16$ plus the Bend constant $b(p)$ of the prime ($b(3) = 7$, $b(5) = 2$, $b(p) = 0$ for $p \geq 7$). Total order- m^2 count $\sum_{p \mid m} b_p$ measured byte-exact at every station, blind-confirmed at $m = 33$ (the line lattice, $3 \mid m$, $11 \mid m$) and at $m = 35$ (a clean Néron-Severi cell, $\gcd(m, 6) = 1$).
- **[The orthogonality — PROVEN]** The discriminant linking form is $(\mathbb{Z}/m)^2$ -equivariant under pencil translation, and pairs a character χ only with its inverse χ^{-1} — of the **same order**. Hence the height-2 part (the order- m^2 torsion) decomposes over character order as an **orthogonal** direct sum of strata, with **zero cross term** between strata of different order. This is structure, not measurement.

- **[The non-separability — PROVEN, the document’s discovery]** Of the three character-order strata of S_m at $p = 3$ (orders $m, 3, m/3$), only the **primitive order- m** stratum carries any height-2 3-torsion. The order-3 stratum is the Fermat cube S_3 , which carries **zero** 3-adic torsion of any height (measured: empty 3-adic profile); the order-5 stratum S_5 carries **no 3-part**. Therefore the entire bend $b(3) = 7$ sits **inside the primitive order- m stratum**: it is **not** a separable sub-Fermat block, **not** piece overlap, **not** a glue leak. The bend is intrinsic to the primitive heart.
- **[The derivation — ROUTE OPEN, honest]** A first-principles derivation of the count law from the three component theorems — a proof of *why* the base and the bend add — remains open. The barrier is now named and is not a gap in effort: the bend does not separate, so no decomposition of the group into a “base block” and a “bend block” can exist. Any future proof must account for the bend *within* the primitive stratum, not as a summand beside it.

Scope. Squarefree odd composite degree. For $\gcd(m, 6) = 1$ the law is read on the Néron-Severi lattice itself (the lines generate NS); for m divisible by 2 or 3 it is read on the SSvL line lattice ($V(2, m)$), the lines’ own span. Prime-power degrees, where the primitive stratum acquires further height (profiles reaching height 3 and 4), are a separate kingdom and are not claimed here.

1. THE THREE RINGS

The name carries two readings, and both are exact.

The three theorems. The composite count is built from three proven laws, each a ring of the chain. The **Watermark** ($|disc\ NS(S_m)| = m^{3(m-3)^2}$ for prime m) fixes the total exponent $a + 2b = 3(m-3)^2$ — the *order*. The **Double Ladder** ($(\mathbb{Z}/m)^a \times (\mathbb{Z}/m^2)^b$ with $b = 3m - 16$ for prime m) fixes the *shape* at prime degree — the base of the count. The **Bend** ($b(p) = 18 - \dim(\text{radical mod } p)$, opening only where $(p-1) \mid k$ for $k \in \{2, 4\}$) fixes the *shift* a dividing prime imposes. The Three-Ring law is the statement that, at composite degree, the shape is the base plus the shift, ring upon ring.

The three primes. Equally, the rings are the prime-types $p \mid m$. At $m = 15 = 3 \cdot 5$ the count splits over the two primes; at $m = 21 = 3 \cdot 7$ over 3 and the flat 7. Each prime contributes its own bend: $p = 3$ bends by 7, $p = 5$ by 2, every $p \geq 7$ is flat. The $p = 5$ and $p = 7$ rings close element-wise — $b(5)$ and $b(7) = 0$ are read directly from the Bend’s power-sum criterion with no further structure. The hard ring is $p = 3$, and it is the hard ring for a reason this document makes precise.

2. THE COUNT LAW — THE FIELD RECORD

The order- m^2 count is measured by the ratified surface manometer (GRAM_MANOMETER_SURFACE_v4.p, the C++ gate `manometer_surface.cpp`), reading the Smith normal form of the SSvL line Gram prime by prime. The law $b_p = (3m - 16) + b(p)$ holds at every station:

- $m = 15, p = 3: 3m - 16 = 29, + b(3) = 7 \rightarrow b_3 = 36$. Measured 36.
- $m = 21, p = 3: 3m - 16 = 47, + b(3) = 7 \rightarrow b_3 = 54$. Measured 54.

- $m = 33$ ($3 \cdot 11$): blind-sealed before computation, confirmed (the line lattice, b_3 and b_{11}).
- $m = 35$ ($5 \cdot 7$): blind-sealed, confirmed — a **clean Néron-Severi cell** ($\gcd(m, 6) = 1$), the law read on NS itself, not the line proxy.

Eight stations, zero off-formula. The two blind confirmations were sealed in the campaign acta with their predicted exponents before the manometer ran, per house protocol. **[FIRM]** — the count is rock; the discriminant group of the composite surface is measured byte-exact.

3. THE ORTHOGONALITY — WHY THERE IS ONLY ONE STRATUM TO MEASURE

The discriminant group of S_m carries an action of the pencil translation torus $(\mathbb{Z}/m)^2$, and the linking form $q : D \times D \rightarrow \mathbb{Q}/\mathbb{Z}$ is invariant under it. By the elementary representation theory of a finite abelian group (Schur), a nondegenerate invariant pairing can couple a character χ only to a character that transforms inversely — χ^{-1} — and a character and its inverse have the **same order**. Consequently the height-2 part of D decomposes over the **order** of the supporting character as an orthogonal sum,

$$D[\text{height } 2] = \oplus_d D_d, \text{ with } q(D_d, D_{\{d'\}}) = 0 \text{ whenever } d \neq d'.$$

The cross term between strata of different order is **identically zero** — a theorem of equivariance, requiring no measurement. This is the lever that turns an intractable count into a single computable question: to find b_3 , one need not understand the whole group, only locate which order-stratum carries the height-2 3-torsion.

For $p = 3 \mid m = 3 \cdot m'$, the relevant strata are three: the **primitive** order- m characters; the order-3 characters, which are pulled back from the Fermat cube S_3 ; and the order- m' characters, pulled back from $S_{\{m'\}}$. By orthogonality,

$$b_3(S_m) = b_3[\text{order } m] + b_3[\text{order } 3] + b_3[\text{order } m'],$$

with no interference. Three terms, cleanly separated by structure. The law's burden reduces to evaluating them.

4. THE NON-SEPARABILITY — THE BEND LIVES IN THE PRIMITIVE

Here the surface answers, and the answer is the discovery.

The order-3 stratum is empty. The order-3 characters are precisely the pull-back of the Fermat cube S_3 along the degree map $m \rightarrow 3$. The cube's own discriminant carries **no 3-adic torsion of any height** — its 3-adic elementary-divisor profile is empty, measured directly and re-measured by the Auditor byte-exact. The proposed “bend block,” the natural candidate for a separable home for $b(3) = 7$, contributes 0.

The order- m' stratum carries no 3-part. At $m = 15$, $m' = 5$: the order-5 stratum is S_5 , whose torsion is 5-adic; it carries no 3-torsion at height 2 (measured 0). At $m = 21$, $m' = 7$: S_7 is flat at 3.

Therefore the bend is intrinsic. With two of the three orthogonal strata contributing nothing,

$$b_3(S_{\{15\}}) = b_3[\text{order } 15] + 0 + 0 = 36,$$

the entire count — base 29 *and* bend 7 together — lives in the **primitive order- m stratum**. The bend $b(3) = 7$ is not a sub-Fermat that can be cut off and counted on its own; it is not the overlap of pieces; it is not a leak of glue across a seam. It is woven into the primitive characters of S_m themselves, indistinguishable by any cut that respects the group's own symmetry. **This is the Three-Ring's proven core, and it is why the count is so much harder to *derive* than to *measure*: there is no seam to derive along.**

5. THE FIVE ROUTES — WHY THE DERIVATION STAYED OPEN

The non-separability theorem was not assumed; it was forced, by the death of every route that presumed a seam. Each route presupposed the bend could be peeled off as a separate object; each died on a number that did not match; and the pattern of the deaths is the proof that the object does not peel.

1. **Singular pieces.** Building the composite group from orbit pieces gave singular Grams. *Autopsy:* bad (imprimitive) orbit representatives — but the fix exposed the next layer.
2. **Φ -stratification.** Saturating each cyclotomic stratum Φ_d separately injected spurious depth $\{3,4\}$. *Autopsy:* stratum-by-stratum saturation is not the geometry; the whole piece is $\{1,2\}$.
3. **The multi-height wall $\{3,4,5,6\}$.** Believed intrinsic; shown an artifact of separate-stratum saturation. The whole un-stratified piece with primitive representatives is clean $(\mathbb{Z}/m)^3 \times (\mathbb{Z}/m^2)^{\{m-4\}}$ — the *flat law of pieces*. No new theory needed.
4. **Sub-degree assembly.** The imprimitive orbits were taken for sub-Fermats at degree m/d . *Autopsy:* the degenerate orbits carry **full** fiber support, not reduced — they are **not** sub-Fermats. The clean-piece criterion is component-wise ($\gcd(a,m)=1$ and $\gcd(b,m)=1$, twenty-one clean at $m=15$), not joint-gcd — a correction the Auditor carried after a dissent-with-data from the Constructor, logged standing.
5. **Overlap, then glue-additivity, then character split.** The clean pieces overcount ($b_{\text{clean}} = 177$ against 36). Read as piece overlap $\rightarrow$ the independent overlap measures 63, never 141 (the 141 was the circular residual $177 - 36$). Read as glue-additivity $b = b_{\text{base}} + b(p)$ with the bend a glue leak $\rightarrow$ the Bend's own ledger records $\text{leak} = 0$; the leak hypothesis is buried in the Bend's graveyard. Read as a clean character split $29 \oplus 7 \rightarrow \S 4$ above: the order-3 block is empty, so the bend block does not exist as a sub-Fermat.

Five disguises of one question — *can the bend be separated?* — and five answers of one word: no. The routes are in the graveyard with pride; each bought the road to $\S 4$.

6. HONEST LEDGER

Proven. The orthogonality of character-order strata (equivariance, Schur); the non-separability of the bend (the order-3 and order- m' strata carry no height-2 3-torsion, so all of b_3 is primitive); the flat law of the whole composite piece ($(\mathbb{Z}/m)^3 \times (\mathbb{Z}/m^2)^{\{m-4\}}$, primitive representatives); the closure of the $p = 5$ and $p = 7$ rings element-wise (the Bend's power-sum criterion); and the instrument — the glue machinery reproduces the prime case ($\alpha = 8/35/125$, $\rho_2 = 8/30/78$, $b = 1/5/17$ at $m = 5, 7, 11$) byte-exact before any composite trust.

Measured [FIRM]. The count law $b_p = (3m - 16) + b(p)$, eight stations, blind at $m = 33$ and the clean cell $m = 35$. The discriminant group order- m^2 count of the composite surface.

[ROUTE OPEN]. A first-principles derivation of the count from the three component theorems. The barrier is characterized, not vague: the bend is non-separable (§4), so the derivation cannot proceed by splitting the group into base and bend; it must account for the bend within the primitive stratum. The validated instrument waits for that route.

Scope honesty. Squarefree odd composite only. Prime-power degree ($m = 9, 25, 27, \dots$) opens further height in the primitive stratum (profiles reaching heights 3, 4) and is a separate kingdom, claimed nowhere here. The $\gcd(m, 6)$ distinction (Néron-Severi for $\gcd(m, 6)=1$, line lattice otherwise) is stated front and center, not folded away.

7. GRAVEYARD, WITH PRIDE

The singular-piece build (bad orbit reps; autopsy yielded the primitivity criterion) · the φ -stratification depth $\{3, 4\}$ (separate-stratum saturation artifact) · the multi-height wall $\{3, 4, 5, 6\}$ (artifact; the whole piece is flat $\{1, 2\}$) · the sub-degree assembly (the degenerate orbits have full support — not sub-Fermats) · the joint-gcd clean-piece criterion (63 against the true component-gcd 21; an Auditor over-generalization from three spot-checks, corrected byte-exact under a Constructor dissent-with-data, logged standing) · the piece-overlap hypothesis (independent overlap $63 \neq 141$; the 141 was circular) · the §132-cyclotomic-direct route alone (controls the order, blind to the glue) · the glue-additivity / glue-leak framing (contradicts the Bend's measured $\text{leak} = 0$, a buried hypothesis) · the character split $29 \oplus 7$ (the order-3 bend block is empty, $0 \neq 7$). *Each grave bought road — and together they are the proof of §4: the bend does not separate, because every seam tried came up empty.*

8. PROVENANCE

Instruments: `GRAM_MANOMETER_SURFACE_v4.py` (ratified), `manometer_surface.cpp` (the gated C++ count), the glue-extraction machinery (validated against the prime case before any composite reading), and the character-stratum profiler (the S_3/S_5 3-adic measurements). Written in crossfire by both instances of the dual protocol and merged under the Auditor's responsibility, with cross-review per house rule; the Auditor re-measured the count and the empty order-3 stratum in a second environment, byte-exact. Seals declared before every measurement; the two blind confirmations ($m = 33$, $m = 35$) sealed in the acta before the manometer ran.

All numbers from files.

References

1. M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963. arXiv:0812.2377. (*The line intersection rules behind the Gram and the shadow map.*)
2. T. Shioda, *Some observations on Jacobi sums*. Adv. Stud. Pure Math. **12** (1987), 119–135. (*The discriminant questions; the composite degrees are the territory this law enters.*)
3. E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (*The published cell-by-cell elementary divisors; the campaign's gates.*)

Companion documents in this repository: THE WATERMARK THEOREM (the prime-degree discriminant order) · THE DOUBLE LADDER THEOREM (the prime-degree group structure, the base $3m-16$) · THE SWEET LIE THEOREM (the fourfold rank) · THE BEND THEOREM (the odd teeth $b(p)$) · THE WILD TOOTH THEOREM (the even half) · CHUCHIPACHI FINDINGS MASTER (the campaign record, §136–§159, graveyards included). The sixth document of the collection — and the first whose headline result is a *non-separability*: the proof that a thing cannot be taken apart. **PMC**.